

International Institute of Information Technology, Hyderabad

MA4.101 – Real Analysis (Monsoon 2026)
Mid Term Examination — Solutions and Grading Rubric

Total Marks: 40 Duration: 90 Minutes

Disclaimer.

1. The solutions given here are not the only ones possible.
2. Results from the book used below are referred to by their numbers in Tao, *Analysis I*: Lemma 3.1.6 (single choice), Proposition 4.3.3 (properties of absolute value and distance), and Proposition 4.4.1 (integer part, which gives the Archimedean property).

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1 Question 1 [10 Marks]

Statement. You may use the following fact without proof: Let $p = (x, y)$ be an ordered pair. Then for all sets X and Y , $p \in X \times Y \iff (x \in X \text{ and } y \in Y)$.

Let A, B, C, D be non-empty sets. Then

- (a) [5 Marks] Show that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$.
- (b) [5 Marks] Show that $A \times B = C \times D$ if and only if $A = C$ and $B = D$.

Solution

Throughout we use the fact stated in the question, which we call (*): for an ordered pair (x, y) and sets X, Y , $(x, y) \in X \times Y \iff (x \in X \text{ and } y \in Y)$. Recall also that, by definition of the Cartesian product, every element of $X \times Y$ is an ordered pair (x, y) with $x \in X$ and $y \in Y$.

(a) ($\Leftarrow$) Suppose $A \subseteq C$ and $B \subseteq D$, and let $p \in A \times B$. Then $p = (x, y)$ with $x \in A$ and $y \in B$. Hence $x \in C$ and $y \in D$, so $p = (x, y) \in C \times D$ by (*). Thus $A \times B \subseteq C \times D$. (This direction does not use non-emptiness.)

($\Rightarrow$) Suppose $A \times B \subseteq C \times D$.

$A \subseteq C$. Let $x \in A$. Since B is non-empty, by the single choice lemma (Lemma 3.1.6) we can choose some $y \in B$. By (*), $(x, y) \in A \times B$, hence $(x, y) \in C \times D$, and by (*) again $x \in C$. As $x \in A$ was arbitrary, $A \subseteq C$.

$B \subseteq D$. Let $y \in B$. Since A is non-empty, choose some $x \in A$. Then $(x, y) \in A \times B \subseteq C \times D$, so $y \in D$ by (*). Hence $B \subseteq D$.

(b) ($\Leftarrow$) If $A = C$ and $B = D$, then $A \times B$ and $C \times D$ are the same set.

($\Rightarrow$) Suppose $A \times B = C \times D$. Then $A \times B \subseteq C \times D$, and since A, B are non-empty, part (a) gives $A \subseteq C$ and $B \subseteq D$. Also $C \times D \subseteq A \times B$, and since C, D are non-empty, part (a) with the roles of the two products exchanged gives $C \subseteq A$ and $D \subseteq B$. Hence $A = C$ and $B = D$, since two sets that are each contained in the other are equal.

Remark. Only $A \neq \emptyset$ and $B \neq \emptyset$ are really needed in (b): if $x \in A$ and $y \in B$, then $(x, y) \in A \times B = C \times D$, so C and D are automatically non-empty. Without the non-emptiness hypothesis both statements fail; for instance $\emptyset \times \{0\} = \emptyset = \emptyset \times \{1\}$ although $\{0\} \neq \{1\}$.

Rubric

Part (a) [5 marks]: $A \times B \subseteq C \times D \iff A \subseteq C \text{ and } B \subseteq D$

Component	Marks
($\Leftarrow$) direction: correct proof that $A \subseteq C$, $B \subseteq D$ implies $A \times B \subseteq C \times D$	2
($\Rightarrow$) direction: correct use of non-emptiness (single choice) to show $A \subseteq C$	1.5
($\Rightarrow$) direction: correct symmetric argument to show $B \subseteq D$	1.5

Part (b) [5 marks]: $A \times B = C \times D \iff A = C \text{ and } B = D$

Component	Marks
$(\Leftarrow)$ direction: correct (trivial) justification	1
$(\Rightarrow)$ direction: correctly applies part (a) to both $A \times B \subseteq C \times D$ and $C \times D \subseteq A \times B$	2
$(\Rightarrow)$ direction: mentions non-emptiness	1
$(\Rightarrow)$ direction: correctly concludes $A = C$ and $B = D$ via mutual containment	1

General notes:

- Important sections under this proof are the decomposition of equality between sets as *Mutual Containment*. For sets X and Y , $X \subseteq Y$ and $Y \subseteq X \iff X = Y$.
- Use of non-emptiness given in the question.
- If a student has provided a new solution, and if it needed the mention of non-emptiness and it is not mentioned, deduct 2/10 marks; similarly, if the use of mutual containment was required and was not used, deduct 3/10 marks. All these deductions are done on a case-by-case basis.

2 Question 2 [10 Marks]

Statement. Define two sequences of natural numbers by $p_1 := 1$, $q_1 := 1$ and

$$p_{n+1} := p_n + 2q_n, \quad q_{n+1} := p_n + q_n.$$

- (a) [5 Marks] Prove that $p_n \geq q_n \geq 1$ for every $n \geq 1$, and that $p_n^2 - 2q_n^2 = (-1)^n$ (an identity between integers).
- (b) [5 Marks] Deduce that $q_{n+1} \geq 2q_n$, hence $q_n \geq 2^{n-1}$, hence $q_n \geq n$ for every $n \geq 1$.

Solution

The first few values are $(p_n, q_n) = (1, 1), (3, 2), (7, 5), (17, 12), (41, 29), \dots$, which is a useful check on everything below.

(a) *The inequalities, by induction on n .* For $n = 1$ we have $p_1 = q_1 = 1$, so $p_1 \geq q_1 \geq 1$. Suppose $p_n \geq q_n \geq 1$ for some $n \geq 1$. Then

$$q_{n+1} = p_n + q_n \geq q_n \geq 1,$$

since adding the natural number p_n cannot decrease q_n . Next,

$$p_{n+1} = p_n + 2q_n = (p_n + q_n) + q_n = q_{n+1} + q_n \geq q_{n+1}.$$

So $p_{n+1} \geq q_{n+1} \geq 1$, which closes the induction.

The identity, by induction on n . The identity is a statement in $\mathbb{Z}$, since its right-hand side may be -1 . For $n = 1$, $p_1^2 - 2q_1^2 = 1 - 2 = -1 = (-1)^1$.

Suppose $p_n^2 - 2q_n^2 = (-1)^n$ for some $n \geq 1$. Expanding,

$$\begin{aligned} p_{n+1}^2 - 2q_{n+1}^2 &= (p_n + 2q_n)^2 - 2(p_n + q_n)^2 \\ &= (p_n^2 + 4p_nq_n + 4q_n^2) - (2p_n^2 + 4p_nq_n + 2q_n^2) \\ &= -p_n^2 + 2q_n^2 = -(p_n^2 - 2q_n^2) = -(-1)^n = (-1)^{n+1}, \end{aligned}$$

using the induction hypothesis in the second-to-last step. This closes the induction.

(b) $q_{n+1} \geq 2q_n$. By (a), $p_n \geq q_n$, and adding q_n to both sides preserves the order, so

$$q_{n+1} = p_n + q_n \geq q_n + q_n = 2q_n.$$

$q_n \geq 2^{n-1}$, *by induction.* For $n = 1$: $q_1 = 1 = 2^0$. If $q_n \geq 2^{n-1}$, then multiplying by 2 preserves the order, so $q_{n+1} \geq 2q_n \geq 2 \cdot 2^{n-1} = 2^n$.

$2^{n-1} \geq n$, *by induction.* For $n = 1$: $2^0 = 1 \geq 1$. If $2^{n-1} \geq n$ for some $n \geq 1$, then

$$2^n = 2 \cdot 2^{n-1} \geq 2n = n + n \geq n + 1,$$

the last step because $n \geq 1$.

Combining the last two facts, $q_n \geq 2^{n-1} \geq n$ for every $n \geq 1$.

Remark. Writing $x_n := p_n/q_n$ and dividing the identity in (a) by q_n^2 gives $x_n^2 - 2 = (-1)^n/q_n^2$, so by (b), $|x_n^2 - 2| = 1/q_n^2 \leq 1/n^2$. The fractions $x_n = 1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \dots$ therefore have squares as close to 2 as we like, although by Proposition 4.4.4 no rational has square exactly 2.

Rubric

Part (a) [5 Marks]

Component	Marks	Criteria
1. Base case (inequalities)	0.5	Checks $p_1 = q_1 = 1$, so $p_1 \geq q_1 \geq 1$.
2. Inductive step: $q_{n+1} \geq 1$	1.0	Assumes $p_n \geq q_n \geq 1$ and shows $q_{n+1} = p_n + q_n \geq q_n \geq 1$, or any equivalent argument.
3. Inductive step: $p_{n+1} \geq q_{n+1}$	1.0	Shows $p_{n+1} = q_{n+1} + q_n \geq q_{n+1}$, or $p_{n+1} - q_{n+1} = q_n \geq 0$, or any equivalent comparison.
4. Base case (identity)	0.5	Verifies $p_1^2 - 2q_1^2 = 1 - 2 = -1 = (-1)^1$.
5. Expansion	1.0	Substitutes the recursions into $p_{n+1}^2 - 2q_{n+1}^2$ and correctly simplifies to $-p_n^2 + 2q_n^2$ or $-(p_n^2 - 2q_n^2)$.
6. Use of induction hypothesis	1.0	Applies $p_n^2 - 2q_n^2 = (-1)^n$ to conclude the expression equals $(-1)^{n+1}$.
Total	5.0	

Marking Guidelines (Part a):

- Proving the inequalities and the identity in a single combined induction is fully acceptable. Award marks for each component wherever it appears.
- If a base case is not stated but is clearly used or obvious from the student's work (e.g., from a table of values), award the base-case mark.
- Checking only the first few values numerically, without an inductive step, earns only the base-case marks (Components 1 and 4).
- Since p_n and q_n are natural numbers, $p_n \geq 0$ may be taken for granted. Do not penalise a student for not justifying this.
- Minor algebraic slips in the expansion that do not change the structure of the argument: deduct at most 0.5 from Component 5, and award Component 6 in full if the induction hypothesis is used correctly.
- If the student writes the result as $-(-1)^n$ and stops without writing $(-1)^{n+1}$, award full marks.
- Any correct alternative proof, such as the matrix form of the recursion or a different algebraic rearrangement, receives full credit.
- If the same error affects more than one step, deduct only once.

Part (b) [5 Marks]

Component	Marks	Criteria
1. $q_{n+1} \geq 2q_n$	1.5	Uses $p_n \geq q_n$ from (a) to get $q_{n+1} = p_n + q_n \geq 2q_n$.
2. Base case for $q_n \geq 2^{n-1}$	0.5	Notes $q_1 = 1 = 2^0$.
3. Inductive step for $q_n \geq 2^{n-1}$	1.0	Shows $q_{n+1} \geq 2q_n \geq 2 \cdot 2^{n-1} = 2^n$.

Component	Marks	Criteria
4. $2^{n-1} \geq n$	1.5	Proves $2^{n-1} \geq n$ for all $n \geq 1$, by induction or any valid argument (0.5 base case, 1.0 inductive step).
5. Conclusion	0.5	Combines the results to conclude $q_n \geq n$ for every $n \geq 1$.
Total	5.0	

Marking Guidelines (Part b):

- A student may use the results of (a) in (b) even if their proof of (a) was incomplete or incorrect. Do not penalise (b) for errors in (a).
- If the student proves $2^{n-1} \geq n$ by another valid method, such as the binomial theorem giving $2^{n-1} = (1+1)^{n-1} \geq 1 + (n-1)$, award full marks for Component 4.
- If the student asserts $2^{n-1} \geq n$ as a well-known fact with some brief justification (e.g., small cases checked plus a remark that 2^{n-1} doubles while n grows by 1), award at least 1.0 out of 1.5 for Component 4.
- If the student proves $q_n \geq n$ directly (e.g., from $q_{n+1} = p_n + q_n \geq q_n + 1$) without going through 2^{n-1} , award full marks for Components 4 and 5, plus whatever their work earns for Components 1–3.
- If the student proves $q_n \geq 2^{n-1}$ by iterating the inequality ($q_n \geq 2q_{n-1} \geq 4q_{n-2} \geq \dots \geq 2^{n-1}q_1$) rather than a formal induction, award full marks for Components 2 and 3.
- If a base case or the final conclusion follows unambiguously from the student's work but is not stated explicitly, award the mark.
- If the same error affects more than one step, deduct only once.
- Award partial credit for any correct method, even if the final result is not reached.

3 Question 3 [10 Marks]

Statement. Let $(a_n)_{n \geq 1}$ be a sequence of rationals. For a given rational $\varepsilon > 0$, the sequence $(a_n)_{n \geq 1}$ is said to be *eventually ε -steady* iff there exists a natural number $N \geq 1$ such that $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq N$.

- (a) [5 Marks] For every rational $\varepsilon > 0$, give an example of an eventually ε -steady sequence of rationals which is not a Cauchy sequence. Justify your answer.
- (b) [5 Marks] Is the sequence $(a_n)_{n \geq 1}$, where $a_n = (-1)^n$ for all $n \geq 1$, a Cauchy sequence? Justify your answer.

Solution

Recall that a sequence is Cauchy if and only if it is eventually δ -steady for every rational $\delta > 0$. So to show a sequence is not Cauchy, it suffices to exhibit one rational $\delta > 0$ for which it is not eventually δ -steady.

(a) Let $\varepsilon > 0$ be an arbitrary rational, and define

$$a_n := (-1)^n \frac{\varepsilon}{2} \quad (n \geq 1),$$

so the sequence is $-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}, -\frac{\varepsilon}{2}, \frac{\varepsilon}{2}, \dots$

Eventually ε -steady. Let $i, j \geq 1$. If i and j have the same parity then $a_i = a_j$ and $|a_i - a_j| = 0 \leq \varepsilon$. If they have different parity then one of a_i, a_j is $\varepsilon/2$ and the other is $-\varepsilon/2$, so $|a_i - a_j| = \varepsilon \leq \varepsilon$. So $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq 1$, and $N := 1$ works.

Not Cauchy. Take $\delta := \varepsilon/2$, a positive rational. Let $N \geq 1$ be arbitrary. The indices $i := N$ and $j := N + 1$ are both at least N and have different parity, so $|a_i - a_j| = \varepsilon > \varepsilon/2 = \delta$. Hence no N works for δ , the sequence is not eventually δ -steady, and so it is not Cauchy.

(b) No. Take $\delta := 1$ and let $N \geq 1$ be arbitrary. With $i := N$ and $j := N + 1$, both at least N ,

$$|a_i - a_j| = |(-1)^N - (-1)^{N+1}| = |(-1)^N| \cdot |1 - (-1)| = 2 > 1.$$

So the sequence is not eventually 1-steady, and hence it is not Cauchy.

Remark. The example in (a) must depend on ε . If a single sequence were eventually ε -steady for every rational $\varepsilon > 0$, it would be Cauchy by definition. For instance $(-1)^n$ from part (b) is eventually ε -steady exactly when $\varepsilon \geq 2$, so it answers (a) only for $\varepsilon \geq 2$. An answer that uses $(-1)^n$ and justifies it only for such ε is incomplete.

Rubric

Part (a) [5 Marks]

Component	Marks	Criteria
1. Correct sequence	2.0	Gives a sequence that is valid for every rational $\varepsilon > 0$ and is intended to satisfy both requirements of the question.
2. Eventually ε -steady	1.5	Correctly proves that there exists $N \geq 1$ such that $ a_i - a_j \leq \varepsilon$ for all $i, j \geq N$.

Component	Marks	Criteria
3. Not Cauchy	1.5	Correctly proves that the sequence is not Cauchy by finding a positive rational δ such that no N satisfies the eventual δ -steadiness condition.
Total	5.0	

Marking Guidelines (Part a):

- If the proposed sequence is **completely invalid** for the given requirement, award 0 for the sequence-construction mark.
- If the proposed sequence is **trivially non-Cauchy and makes no genuine attempt to satisfy the eventual ε -steadiness requirement**, award 0 for the non-Cauchy argument as well. If the student has made a **genuine attempt to satisfy the eventual ε -steadiness requirement**, even if the proposed sequence is ultimately incorrect, the non-Cauchy marks may be awarded when the non-Cauchy argument for their proposed sequence is mathematically correct.
- If the proposed sequence is **trivially eventually ε -steady and makes no genuine attempt to show that it is not Cauchy**, award 0 for the eventual ε -steadiness argument as well. If the student has made a **genuine attempt to show that the sequence is not Cauchy**, even if the non-Cauchy argument is ultimately incorrect, the eventual ε -steadiness marks may be awarded when the sequence is correctly shown to be eventually ε -steady.
- If the same mathematical error affects more than one step, do not deduct marks repeatedly for that single error. However, a student should not receive marks for proving a property that is false for their proposed sequence. Marks should be awarded only for properties that actually hold for the sequence they have chosen and meet the requirements.
- If the proposed sequence is not completely invalid but contains a minor error in its construction, assess the sequence and the subsequent arguments according to the extent to which the required properties are still correctly established; award partial marks at relevant places.

Part (b) [5 Marks]

Component	Marks	Criteria
1. Identify the answer	0.5	Correctly states that the sequence is not Cauchy .
2. Choose a suitable tolerance	1.0	Chooses any positive rational δ for which the Cauchy condition can be shown to fail.
3. Show the failure occurs arbitrarily far out	1.5	Takes an arbitrary N and identifies two indices $i, j \geq N$ whose corresponding terms are different (e.g., consecutive terms or any two indices of opposite parity).
4. Show the terms remain too far apart	1.5	Correctly establishes that the distance between the chosen terms is 2, and hence is greater than the chosen δ . Equivalent correct calculations or arguments should receive full credit.

Component	Marks	Criteria
5. Connect to the definition of Cauchy	0.5	Explains that since for every N there are $i, j \geq N$ with $ a_i - a_j > \delta$, the sequence is not eventually δ -steady and therefore is not Cauchy.
Total	5.0	

Marking Guidelines (Part b):

- If the student initially states the wrong conclusion but their subsequent argument correctly proves that the sequence is not Cauchy, award marks based on the mathematical work and do not penalize the same error again.
- If the student states the correct conclusion without providing a valid argument, award only the marks supported by the work shown.
- If the student's choice of δ is invalid and is never corrected, the corresponding mark should not be awarded. Subsequent work that depends on this invalid choice should be assessed on its mathematical validity rather than automatically receiving full credit.
- If the student uses a valid value of δ but makes a minor computational or comparison error, deduct only for the affected calculation. Do not penalize the same error again in later steps.
- If the student's chosen indices do not satisfy the required conditions, deduct the marks associated with that choice. If the remainder of the argument is independently correct, award the corresponding remaining marks.
- If a missing justification can be inferred unambiguously from the student's work, do not penalize the student merely for not explicitly stating it.
- If an earlier error leads to an incorrect later expression, but the student applies the correct method consistently to their earlier result, award partial credit for the valid method demonstrated.
- If the student's argument is fundamentally incomplete or based on an incorrect use of the definition, award only the marks for independently correct mathematical work.

4 Question 4 [10 Marks]

Statement. A sequence $(a_n)_{n \geq 1}$ of rationals is called *anchored* if for every rational $\varepsilon > 0$ there exists natural $N \geq 1$ with

$$|a_j - a_N| \leq \varepsilon \quad \text{for all } j \geq N.$$

Prove that a sequence of rationals is anchored if and only if it is a Cauchy sequence.

Solution

($\Leftarrow$) *Cauchy implies anchored.* Let (a_n) be Cauchy and let $\varepsilon > 0$ be rational. By definition there is $N \geq 1$ with $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq N$. Taking $i := N$, which satisfies $N \geq N$, gives $|a_j - a_N| \leq \varepsilon$ for all $j \geq N$. So (a_n) is anchored.

($\Rightarrow$) *Anchored implies Cauchy.* Let (a_n) be anchored and let $\varepsilon > 0$ be rational. Then $\varepsilon/2$ is also a positive rational, so by the anchored property there is $N \geq 1$ with

$$|a_j - a_N| \leq \frac{\varepsilon}{2} \quad \text{for all } j \geq N.$$

Now let $i, j \geq N$. By the triangle inequality (Proposition 4.3.3),

$$|a_i - a_j| = |(a_i - a_N) + (a_N - a_j)| \leq |a_i - a_N| + |a_j - a_N| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where we used $|a_N - a_j| = |a_j - a_N|$. So (a_n) is eventually ε -steady. As ε was an arbitrary positive rational, (a_n) is Cauchy.

Remark. The anchored property must be applied with $\varepsilon/2$, not ε . Applying it with ε only gives $|a_i - a_j| \leq 2\varepsilon$, which does not match the definition of a Cauchy sequence without a further argument.

Rubric

Direction ($\Leftarrow$): Cauchy implies anchored [5 Marks]

Component	Marks	Criteria
1. Setup	1.0	An arbitrary rational $\varepsilon > 0$ is taken (not a specific value).
2. Use of the Cauchy definition	1.0	The definition of a Cauchy sequence is correctly applied to obtain $N \geq 1$ with $ a_i - a_j \leq \varepsilon$ for <i>all</i> $i, j \geq N$.
3. Substitution $i := N$	2.0	The value $i := N$ is substituted (valid since $N \geq N$), giving $ a_j - a_N \leq \varepsilon$ for all $j \geq N$. 1.0 mark is awarded if the substitution is made but it is not made clear that the bound holds for every $j \geq N$.
4. Conclusion	1.0	This N is identified as a witness for the anchored condition, and, since ε was arbitrary, the sequence is concluded to be anchored.
Total	5.0	

Direction ($\Rightarrow$): Anchored implies Cauchy [5 Marks]

Component	Marks	Criteria
1. Choice of $\varepsilon/2$	1.0	An arbitrary rational $\varepsilon > 0$ is taken, and the anchored property is applied with $\varepsilon/2$ (or another suitable fraction such as $\varepsilon/3$), which is noted to be a positive rational.
2. Obtaining N	1.0	The resulting $N \geq 1$ with $ a_j - a_N \leq \varepsilon/2$ for all $j \geq N$ is correctly obtained from the anchored property.
3. Triangle inequality	2.0	For arbitrary $i, j \geq N$, the difference is decomposed as $a_i - a_j = (a_i - a_N) + (a_N - a_j)$ and bounded by $\varepsilon/2 + \varepsilon/2 = \varepsilon$, with the symmetry $ a_N - a_j = a_j - a_N $ used explicitly or implicitly. 1.0 mark is awarded if the decomposition is correct but the bound is incomplete or incorrect.
4. Conclusion	1.0	The sequence is concluded to be eventually ε -steady and, since ε was arbitrary, Cauchy.
Total	5.0	

Marking Guidelines:

- **Anchored property applied with ε instead of $\varepsilon/2$.** If only $|a_i - a_j| \leq 2\varepsilon$ is obtained and the sequence is then declared Cauchy, Components 1 and 4 of Direction ($\Rightarrow$) are not awarded (maximum 3.0 out of 5 for that direction). If a valid correction is added (for example, the argument is repeated with $\varepsilon/2$ in place of ε , since ε was arbitrary), full marks are awarded.
- **Quantifier errors.** If N is allowed to depend on j , if ε is taken to be a specific number, or if the bound is proved only for some i, j rather than for all $i, j \geq N$, the affected direction is capped at 3.0 out of 5.
- **Only $|a_j - a_N| \leq \varepsilon$ shown in Direction ($\Rightarrow$).** This merely restates the hypothesis, as $|a_i - a_j|$ is never bounded for general i, j . At most 2.0 out of 5 (Components 1 and 2) is awarded for that direction.
- **Appeals to limits or convergence.** No credit is given for the affected direction if it is assumed that a Cauchy sequence of rationals converges (to a rational or a real number). This is false in $\mathbb{Q}$, and circular if the reals have not yet been constructed from Cauchy sequences.
- **No deduction** is made if:
 - strict inequalities ($<$) are used instead of $\leq$;
 - Proposition 4.3.3 is not cited by number;
 - the rationality of $\varepsilon/2$ is not remarked upon, provided everything else is correct;
 - the two directions are proved in the opposite order, or ($\Leftarrow$) and ($\Rightarrow$) are labelled the other way round, provided both proofs are correct.
- **Overall.** A complete and correct proof of one direction, with nothing written for the other, is awarded 5.0 out of 10.

5 Question 5 [10 Marks]

Statement. A function $D : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ is said to be a *distance measure* on $\mathbb{Q}$ if for all $a, b, c \in \mathbb{Q}$:
 (1) $D(a, b) \geq 0$ and equality holds iff $a = b$. (2) $D(a, b) = D(b, a)$. (3) $D(a, b) \leq D(a, c) + D(c, b)$.
 A sequence $(a_n)_{n \geq 1}$ of rationals is called *D-Cauchy* iff

$$\forall \text{ rational } \varepsilon > 0 \exists \text{ natural } N \geq 1 \text{ such that } D(a_i, a_j) \leq \varepsilon \forall i, j \geq N.$$

Then for $D(a, b) = |a - b|$, we get the usual definition of Cauchy sequences. Now define a function $\phi : \mathbb{Q} \rightarrow \mathbb{Q}$ by $\phi(x) = 1/x$ for all nonzero rationals x and $\phi(0) = 0$. Then define $D_\phi(a, b) = |\phi(a) - \phi(b)|$ for all $a, b \in \mathbb{Q}$.

- (a) [5 Marks] Prove that D_ϕ is a valid distance measure on $\mathbb{Q}$.
 (b) [5 Marks] Give an example of a sequence of rationals that is D_ϕ -Cauchy but not a Cauchy sequence. Justify your answer.

Solution

(a) First, ϕ is a function from $\mathbb{Q}$ to $\mathbb{Q}$: for $x \neq 0$ the reciprocal $1/x$ is a rational number, and $\phi(0) = 0$. Hence $D_\phi(a, b)$ is a rational number for all $a, b \in \mathbb{Q}$.

ϕ is injective. Suppose $\phi(a) = \phi(b)$. We first note that $\phi(x) \neq 0$ whenever $x \neq 0$: if $1/x = 0$ then $1 = x \cdot (1/x) = x \cdot 0 = 0$, which is false. Now consider cases.

- If $a = 0$ and $b = 0$, then $a = b$.
- If exactly one of a, b is 0, say $a = 0 \neq b$, then $\phi(a) = 0 \neq \phi(b)$, contradicting $\phi(a) = \phi(b)$. So this case cannot occur.
- If $a \neq 0$ and $b \neq 0$, then $1/a = 1/b$; multiplying both sides by ab gives $b = a$.

In every possible case $a = b$.

We now verify the three properties, writing $x := \phi(a)$, $y := \phi(b)$, $z := \phi(c)$.

(1) $D_\phi(a, b) = |x - y| \geq 0$ by Proposition 4.3.3(a). Moreover $|x - y| = 0$ if and only if $x = y$, again by Proposition 4.3.3(a). Now $x = y$ means $\phi(a) = \phi(b)$, which holds if and only if $a = b$: the "if" direction is clear, and the "only if" direction is injectivity of ϕ . So $D_\phi(a, b) = 0$ if and only if $a = b$.

(2) $D_\phi(a, b) = |x - y| = |-(y - x)| = |y - x| = D_\phi(b, a)$, using $|-w| = |w|$ from Proposition 4.3.3(d).

(3) By the triangle inequality for distance, Proposition 4.3.3(g),

$$D_\phi(a, b) = |x - y| \leq |x - z| + |z - y| = D_\phi(a, c) + D_\phi(c, b).$$

So D_ϕ is a distance measure on $\mathbb{Q}$.

(b) Let $a_n := n$ for $n \geq 1$. Since $n \neq 0$, $\phi(a_n) = 1/n$, so

$$D_\phi(a_i, a_j) = \left| \frac{1}{i} - \frac{1}{j} \right|.$$

(a_n) is D_ϕ -Cauchy. Let $\varepsilon > 0$ be rational. By the Archimedean property (Proposition 4.4.1) there is an integer M with $M \leq 1/\varepsilon < M + 1$. Since $1/\varepsilon > 0$ we have $M + 1 > 0$, so $N := M + 1$ is a natural number with $N \geq 1$ and $N > 1/\varepsilon$. Multiplying by the positive rational ε/N gives $1/N < \varepsilon$.

Now let $i, j \geq N$. By symmetry we may assume $i \leq j$. Then $0 < 1/j \leq 1/i \leq 1/N$, so

$$D_\phi(a_i, a_j) = \frac{1}{i} - \frac{1}{j} \leq \frac{1}{i} \leq \frac{1}{N} < \varepsilon.$$

Hence (a_n) is D_ϕ -Cauchy. (Equivalently: $(1/n)$ is a Cauchy sequence, Proposition 5.1.11 of Tao, and $D_\phi(a_i, a_j)$ is exactly $|1/i - 1/j|$.)

(a_n) is not Cauchy. Take $\delta := 1/2$ and let $N \geq 1$ be arbitrary. With $i := N + 1$ and $j := N$, both at least N ,

$$|a_i - a_j| = |(N + 1) - N| = 1 > \frac{1}{2}.$$

So (a_n) is not eventually $\frac{1}{2}$ -steady, and hence not Cauchy.

Remark. Two further observations. First, $D_\phi(n, 0) = |1/n - 0| = 1/n$, so measured with D_ϕ the sequence $1, 2, 3, \dots$ approaches 0. Second, the implication also fails the other way for the same distance: $(1/n)$ is Cauchy, but $D_\phi(1/i, 1/j) = |i - j|$, so it is not D_ϕ -Cauchy. Whether a sequence is Cauchy depends on how distance is measured, not only on the sequence itself; the real numbers are built from the particular choice $D(a, b) = |a - b|$.

Rubric

Part (a) [5 Marks]

Criteria	Marks	Description & Partial Credit
Setup & Well-definedness	1.0	1.0 mark for explicitly showing that $D_\phi(a, b)$ maps to $\mathbb{Q}$, which is a core requirement for it to be a distance measure on $\mathbb{Q}$. Broken down as follows: • 0.5 marks for the non-zero case: Expressing $D_\phi(a, b)$ using the definition of ϕ (e.g., $1/a - 1/b$) and specifically stating/concluding that this evaluates to a rational number. • 0.5 marks for the zero case: Explicitly handling/noting $\phi(0) = 0$ to show the function is fully well-defined over all of $\mathbb{Q}$.
Property 1: Positivity & Equality	2.0	0.5 marks for showing non-negativity: $D_\phi(a, b) = \phi(a) - \phi(b) \geq 0$. 1.5 marks for proving $D_\phi(a, b) = 0 \iff a = b$, broken down as follows: • 0.5 marks for the basic setup: $\phi(a) - \phi(b) = 0 \iff \phi(a) = \phi(b)$. • 1.0 mark for providing mathematical support showing that $\phi(a) = \phi(b) \implies a = b$. (Students do not need to explicitly use the word "injective" as long as they show the mathematical steps/logic proving that no two different inputs yield the same output.)

Criteria	Marks	Description & Partial Credit
Property 2: Symmetry	1.0	1.0 mark for showing $D_\phi(a, b) = D_\phi(b, a)$ by correctly using the symmetric property of standard absolute values ($ x - y = y - x $), with the case where a or b is 0 covered (either handled explicitly, or covered automatically by working with $\phi(a)$ and $\phi(b)$ directly). Partial Credit: 0.5 marks if the argument only treats non-zero a, b (e.g., written only as $ 1/a - 1/b = 1/b - 1/a $) and the case where a or b is 0 is not handled.
Property 3: Triangle Inequality	1.0	1.0 mark for correctly applying the standard triangle inequality to ϕ : $D_\phi(a, b) = \phi(a) - \phi(b) \leq \phi(a) - \phi(c) + \phi(c) - \phi(b) = D_\phi(a, c) + D_\phi(c, b)$, with the case where any of a, b, c is 0 covered (either handled explicitly, or covered automatically by working with $\phi(a), \phi(b), \phi(c)$ directly). Partial Credit: 0.5 marks if the argument only treats non-zero a, b, c (e.g., written only in terms of $1/a, 1/b, 1/c$) and the case where one of them is 0 is not handled.
Total	5.0	

Part (b) [5 Marks]

Note: Students are not restricted to the sequence $a_n = n$. Any rational sequence (a_n) that diverges/does not converge in $\mathbb{R}$, but for which the reciprocal sequence converges to 0, is a valid counterexample—e.g., $a_n = n^2$, $a_n = 2n$, or $a_n = (-1)^n n$.

Criteria	Marks	Description & Partial Credit
Valid Sequence Example	1.0	1.0 mark for explicitly defining a valid sequence of rationals (a_n) where $a_n \not\rightarrow L$ but $\phi(a_n) \rightarrow 0$. 0 marks if the sequence is standard Cauchy or if $\phi(a_n)$ is not Cauchy.
Justification 1: Is D_ϕ -Cauchy	2.0	2.0 marks for rigorously proving that the chosen sequence is D_ϕ -Cauchy (i.e., proving $\phi(a_n)$ is a standard Cauchy sequence using the formal definition $\forall \epsilon > 0, \exists N \dots$). Partial Credit: 1.0 mark for evaluating $\phi(a_n)$ and intuitively explaining it gets arbitrarily close to 0, but lacking formal ϵ - N rigor.

Criteria	Marks	Description & Partial Credit
Justification 2: Is NOT Cauchy	2.0	2.0 marks for rigorously proving the sequence is not a standard Cauchy sequence. Typically done by showing $\exists \epsilon > 0$ such that for any N , we can find $i, j \geq N$ where $ a_i - a_j \geq \epsilon$. Partial Credit: 1.0 mark for stating the sequence diverges to infinity without formal bounds, or for setting up the negation of the Cauchy definition but failing to execute it properly.
Total	5.0	